

CHARLES SHALUA

Software Engineer — AI & Backend | Founder, Data-Life Tech

Toronto, ON · data-life.tech · github.com/charlesAlgo · linkedin.com/in/charles-shalua

Professional Summary

AI and backend engineer who has shipped production software into a live business. Built and released an AI-assisted estimating and operations platform for a commercial electrical contractor — spanning a vision-model document-extraction pipeline, a shared cloud database, and a packaged desktop application delivered to v1.0.0 and installed on working machines. Founder of Data-Life Tech, delivering document automation, RAG systems, and workflow integration to small and mid-sized businesses. Strong at translating technical systems for non-technical stakeholders, and disciplined about the part of AI work that decides whether it actually gets adopted: human review, explainability, and trust.

Core Skills

AI & LLM Engineering — RAG pipelines, vision-language document extraction, structured output (JSON mode), prompt engineering, model evaluation, human-in-the-loop system design, Model Context Protocol (MCP), local model deployment (Ollama), API cost modelling and tracking

Backend & Data — Python, SQL, PostgreSQL, Supabase, REST API design and integration, transactional data integrity, schema design, data pipelines, secrets management

Software Engineering — JavaScript, Git/GitHub, code review, debugging and root-cause analysis, release engineering and packaging, technical documentation, Agile collaboration

Analytics — Power BI, Pandas, NumPy, KPI dashboards, Excel

Professional Experience

Data-Life Tech — Founder & AI Engineer

Toronto, ON · 2023 – Present

Independent AI and automation practice serving small and mid-sized businesses. Services delivered:

- **Document automation** — converting PDFs, drawings, and forms into structured, priced data with a human approval step before anything is used
- **RAG and knowledge systems** — retrieval over client documents with source-grounded LLM responses
- **Workflow and API integration** — connecting business tools to remove repetitive manual data entry
- **Analytics and reporting** — Power BI dashboards for KPI tracking and executive reporting
- **Web platforms** — designed and launched data-life.tech and client-facing sites

Scope engagements directly with business owners, translating operational pain points into delivery plans. Built reusable extraction and retrieval components, with per-call cost logging so work is priced on real usage rather than estimates.

Alton Electric Contracting Inc. — Software Engineer, AI & Backend - Internship

Oshawa, ON · June 2026 – September 2026

- Led an **AI-assisted PDF takeoff pipeline** that reads construction drawings and produces estimator-reviewed line items, removing hours of manual symbol counting and data entry per bid
- Designed a **shared cloud database** (Supabase / PostgreSQL) as the single source of truth for estimating data, replacing fragmented local files across the team
- Removed hardcoded credentials from the codebase and migrated secrets and file storage to managed cloud services
- Audited and completed the estimating module, resolving concurrency and state-synchronisation defects that threatened data integrity
- Owned **packaging and release engineering** through to an installed v1.0.0 application running on company machines
- Built the shared development environment and onboarded teammates; reviewed code and drove fixes to root causes rather than symptoms

"They treated this as production software, not a class project, and the result showed it." — Densley Thomas, Owner, Alton Electric Contracting Inc.

FlyRank — Backend AI Engineer - Internship

Remote · June, 2026 - Present

- Backend AI engineering on FlyRank's platform.

Tanzania Telecommunication Company Limited (TTCL) — IT Support Specialist

Dec 2019 - May 2022

- Diagnosed and resolved hardware, software, and network faults across enterprise systems supporting hundreds of users
- Delivered hands-on technical training to end users, improving system adoption and reducing repeat support tickets
- Contributed to system upgrades and infrastructure monitoring, strengthening security posture and reliability

Earlier Roles

Atlantic Packaging — Shipper & Receiver Clerk (Nov 2024 - Dec 2025) · Olymel — Production Machine Operator (Jul 2023 - Aug 2024). High-accuracy documentation and process discipline in regulated, high-volume environments.

Education

Centennial College — Software Engineering & Artificial Intelligence · 2025 - Present

Certifications

Claude Code in Action Certificate — Anthropic (Completed)

Google Data Analytics Professional Certificate — Coursera (Completed)

AI / LLM Engineering — Advanced Certification (In Progress)

Advanced Data Analytics — Professional Certificate (In Progress)

Core Strengths

Analytical problem solving · Systems thinking · Ownership and accountability · Stakeholder communication · Continuous learning